

Eurobioc2026 BiocContainer Report

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Introduction

Containers enhance reproducibility and additionally offer a convenient way to share the setup used and needed to run a script. Offering a way to create containers which contain desired R packages from within R allows users to more easily create these containers and work with them. R packages like `dockerfiler` Fay et al. (2026) offer functions to create and work with Docker files within R. However, currently there is no convenient way to create and share Docker files and containers from R while also controlling for the Bioconductor release and including data or scripts. The aim in this hackathon was to create a minimal set of functions that offer this possibility, and this work eventually gave rise to the R package `SimpleBiocContainer`.

R package `SimpleBiocContainer`

As part of the European Bioconductor conference EuroBioc2026 Hackathon, we created an R package called `SimpleBiocContainer` which offers a more user friendly way to create, build and push Docker containers to [Dockerhub](#) from R. A working version of the package can be found on GitHub: <https://github.com/lgatto/SimpleBiocContainer.git>.

The functions in `SimpleBiocContainer` make use of the existing Bioconductor docker containers (see this [link](#) for more details), which already contain system dependencies and a few basic packages. On top of that, we add user-specified Bioconductor/R packages using `BiocManager::install()`. The Bioconductor version, and thus the corresponding docker container, is automatically determined from the R session.

Currently, the package has two exported functions:

- `makeSimpleBiocContainer()` creates a container directory and populates it with a Dockerfile and optional data and script directories with corresponding user-defined files.
- `buildPushDocker()` builds and pushes the container to [Dockerhub](#) or optionally [Github container registry](#) directly from R.

Here is a simple example to illustrate its use:

```
library(SimpleBiocContainer)

container_path <- makeSimpleBiocContainer(
  package = "MsCoreUtils",
  container = "my_container",
  data = "data.rda",
  script = "script.R")
```



```
buildPushDocker(  
  container = container_path,  
  dockerUsername = "docker_username")
```

Conclusions / Future Work

The current version of the package covers all features that we wanted to include, but represents a simple first implementation targeting users without or with limited Docker experience.

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References

Fay, C., Guyader, V., Parry, J., & Rochette, S. (2026). *Dockerfiler: Easy dockerfile creation from r*. <https://doi.org/10.32614/CRAN.package.dockerfiler>

Author contributions

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